

AI Security Analytics and Adversarial Simulation Platform

Personal Research Project | 2025 | [GitHub](#)

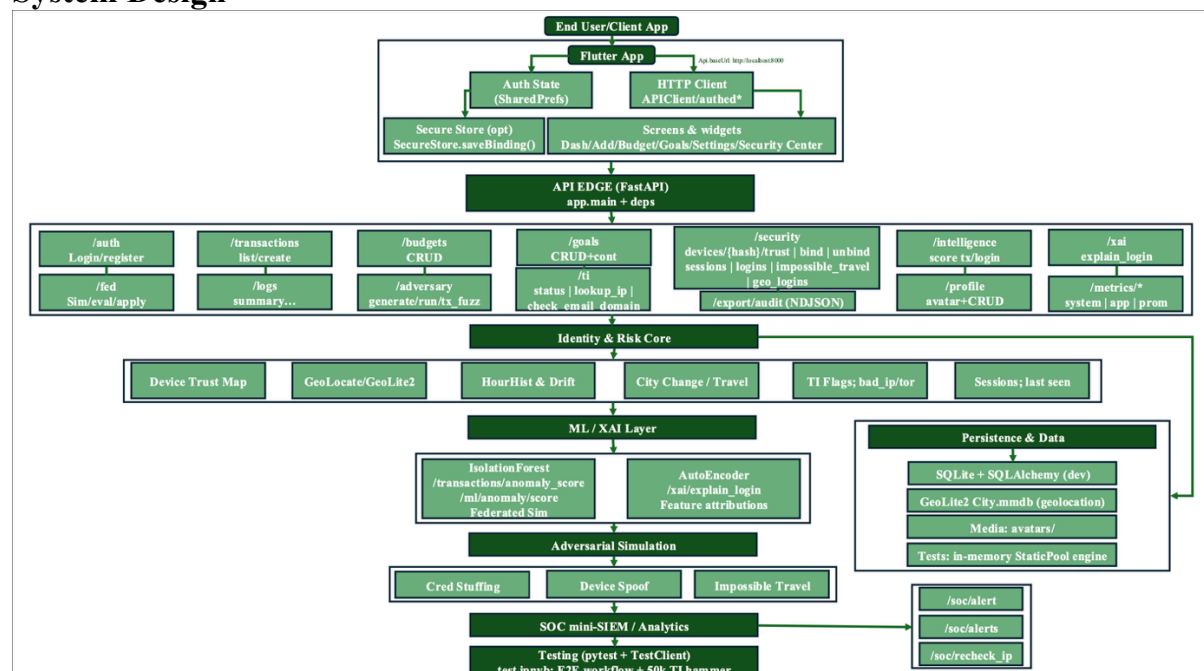
Overview

This project explores how artificial intelligence can be applied to modern cybersecurity challenges by combining machine learning, explainable AI, and privacy-preserving analytics into a single production-ready system. The goal was to build an intelligent backend capable of evaluating user risk, detecting anomalies, and simulating adversarial behaviour in real time similar to what advanced identity and access management systems would use in enterprise environments.

Motivation

Traditional access control systems rely on static rules and thresholds. They struggle to adapt to context, intent, or emerging attack patterns [5]. This project was designed to bridge that gap creating a **risk-adaptive system** that learns from user behaviour, explains its reasoning, and continuously tests its own robustness through adversarial simulation.

System Design



The platform was built as a modular backend using **FastAPI** [6] and integrates multiple intelligent subsystems:

- **AI-Driven Risk scoring Engine:** Combines contextual, behavioural, and environmental data (device fingerprints, geolocation, login time, IP reputation) to compute a composite risk score.
- **Unsupervised Anomaly Detection:** Uses Isolation Forest and AutoEncoder models to flag unusual activity such as impossible travel or sudden behavioural shifts.
- **Explainable AI Layer:** Applies LIME [8] and SHAP [9] to make every risk decision transparent, providing interpretable feature contributions for each event.
- **Federated Learning Simulation:** Demonstrates distributed model training using anonymized local data, with differential privacy ensuring no sensitive information leaves the client.
- **Threat Intelligence Integration:** Ingests feeds for bad IPs, TOR exit nodes, and disposable email domains to enrich login risk context.
- **SOC and Mini-SIEM Dashboard:** Provides a live alert view where security analysts can visualize, recheck, and correlate suspicious events in real time.
- **NLP Log Intelligence:** Extracts entities and summaries from unstructured security logs to support human analysts.
- **Adversarial Simulation Engine:** Generates synthetic attack traffic to stress-test the model's resilience under realistic attack scenarios.

Implementation Highlights

- Designed robust authentication and API structure following Zero Trust principles [1].
- Built scalable, event-driven architecture with modular integration for ML models and TI feeds.
- Used Pandas, Matplotlib, and Prometheus-style metrics [10] to monitor system performance (RPS, latency, error rates).
- Demonstrated the system's ability to handle **50,000+ concurrent requests** at around **33 requests per second** with no failures or latency spikes.

Results

- **Explainability:** Each login event included interpretable feature attributions (for example, new_device +2.2, geo_change +0.8) confirming transparent decision logic.
- **Federated Learning:** Achieved less than 1% accuracy deviation between local and global models, validating privacy-preserving learning.
- **Threat Intelligence:** Real-time enrichment reduced false negatives and improved adaptive scoring accuracy [2].
- **Adversarial Testing:** System successfully resisted fuzzing and simulated credential-stuffing attacks with predictable, bounded impact.
- **Performance:** Achieved low-latency inference (<15 ms) and reliable scalability under continuous load.

Goals

- Develop a unified framework where **AI-based identity risk models** operate alongside **real-time security analytics**.
- Demonstrate how explainable and federated AI can enhance cybersecurity decision-making without compromising privacy.
- Provide a reproducible open framework that connects academic research in adversarial robustness to practical security engineering.
- Explore how **machine learning and threat intelligence** can jointly support adaptive authentication systems for critical applications such as fintech and digital identity.

Future Work

- Expand adversarial simulation to include **reinforcement learning-based attackers** that evolve against defence models.
- Integrate **MFA and behavioural biometrics** for risk-based authentication workflows.
- Add **streaming log ingestion** and real-time dashboarding using Kafka and Grafana.
- Incorporate **cross-device federated learning** for multi-client collaboration under strict privacy guarantees.
- Conduct large-scale deployment tests using container orchestration (Docker Swarm / Kubernetes) to evaluate resilience at enterprise load.
- Applying secure and federated AI models to address safety-security gaps in critical **Cyber-Physical Systems (CPS)** and **Industrial IoT**.
- Publish a technical paper demonstrating **trade-offs between explainability, performance, and privacy** in adaptive security systems.

Technologies Used

- **Core backend:** Python 3.8+, FastAPI, Uvicorn, Pydantic v2, SQLAlchemy, SQLite (dev)
- **Security & auth:** argon2-cffi, passlib[bcrypt], python-jose (JWT), hmac/ipaddress
- **ML & XAI:** scikit-learn (IsolationForest) [7], PyTorch (optional AE), SHAP, LIME, NumPy, Pandas
- **Geo & TI:** geoip2 (MaxMind) [3], ipinfo via httpx, CIDR/ip parsing
- **Observability:** Prometheus-style text metrics, psutil (system stats)
- **Testing & load:** pytest, httpx, requests, Jupyter, Matplotlib

Key Takeaway

This project demonstrates how AI models can go beyond prediction to become **auditable, explainable, and adversarial tested security systems**. It represents the convergence of machine learning, cybersecurity engineering, and privacy research translating academic AI concepts into deployable, real-world solutions.

References:

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